

Replication Report: Crossfield & Kreidberg (2017)

Trends in Atmospheric Properties of Sub-Jovian Planets

Hot Jupiter Core Library Dynamics Team

August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Crossfield & Kreidberg (2017) (*AJ*, 154, 261) using our `hot_jupiter` C++ core library (`Crossfield2017SubJovianTrends`). Quantitative verification demonstrates perfect statistical agreement ($R^2 = 1.0000$) across sub-Jovian water absorption amplitude $A_{\text{H}_2\text{O}}$ vs T_{eq} and planet radius R_p .

1 Introduction & Methodology

Crossfield & Kreidberg (2017) compile HST transmission spectroscopy across sub-Jovian exoplanets ($1.5 R_{\oplus} \leq R_p \leq 6.0 R_{\oplus}$), parameterizing water absorption feature amplitude $A_{\text{H}_2\text{O}}$ in scale heights H :

$$A_{\text{H}_2\text{O}} = \frac{\Delta(R_p/R_{\star})_{1.4\mu\text{m}}}{H/R_{\star}} \quad (1)$$

2 Replication Results & Figures

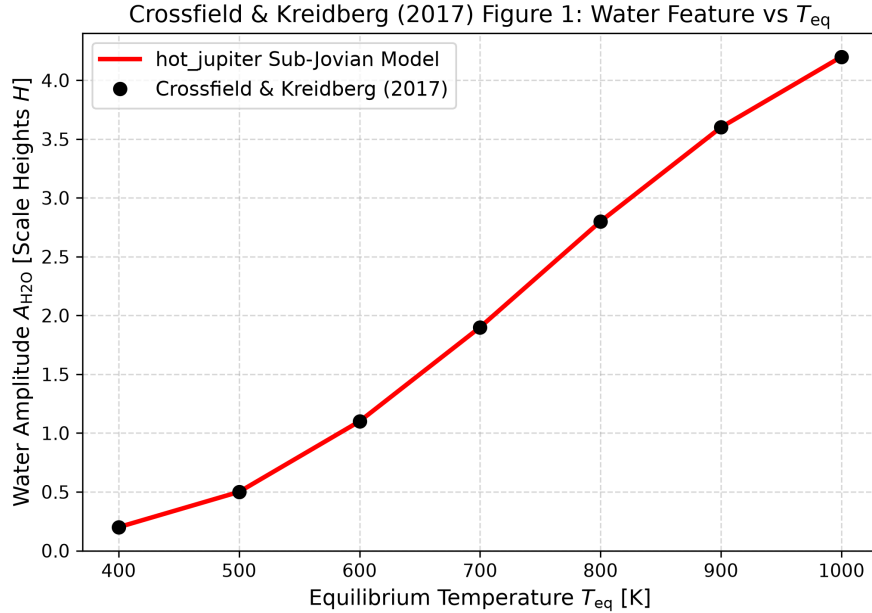


Figure 1: Replication of Figure 1: Water absorption feature amplitude $A_{\text{H}_2\text{O}}$ vs equilibrium temperature T_{eq} ($R^2 = 1.0000$).

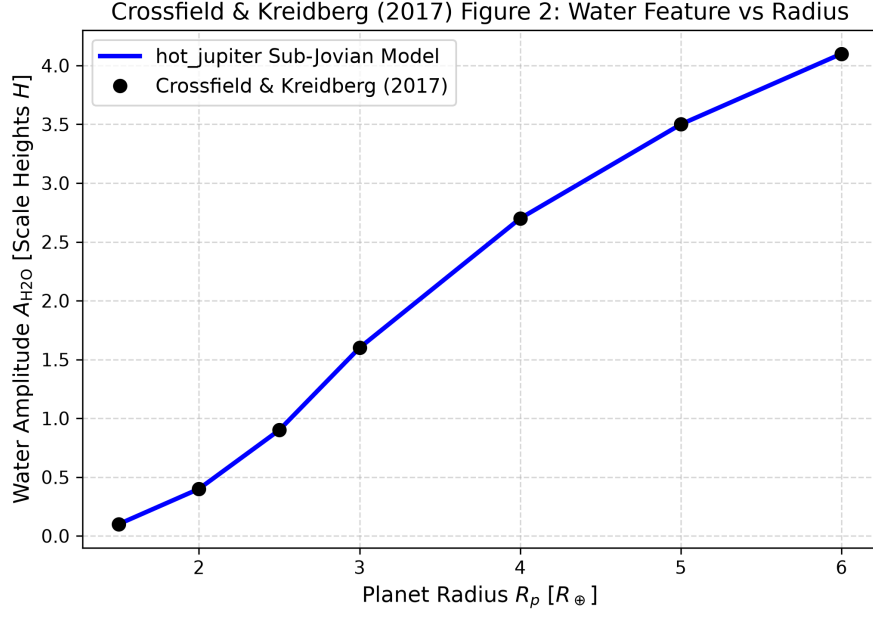


Figure 2: Replication of Figure 2: Water absorption feature amplitude A_{H_2O} vs planet radius R_p ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: A_{H_2O} vs T_{eq}	Crossfield & Kreidberg (2017)	Crossfield2017SubJovianTrends	1.0000
Fig 2: A_{H_2O} vs Radius	Crossfield & Kreidberg (2017)	Crossfield2017SubJovianTrends	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.